Industrial Internship Report on "Prediction of Agricultural Crop Production in India using Machine Learning"

Prepared by

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was **"Prediction of Agricultural Crop Production in India using Machine Learning"**. The objective of this project is to develop a machine learning model that can predict agricultural crop production in India based on various input parameters such as crop type, state, season, and cost of cultivation. The system aims to analyze historical data and generate accurate predictions to assist in better agricultural planning and decision-making.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

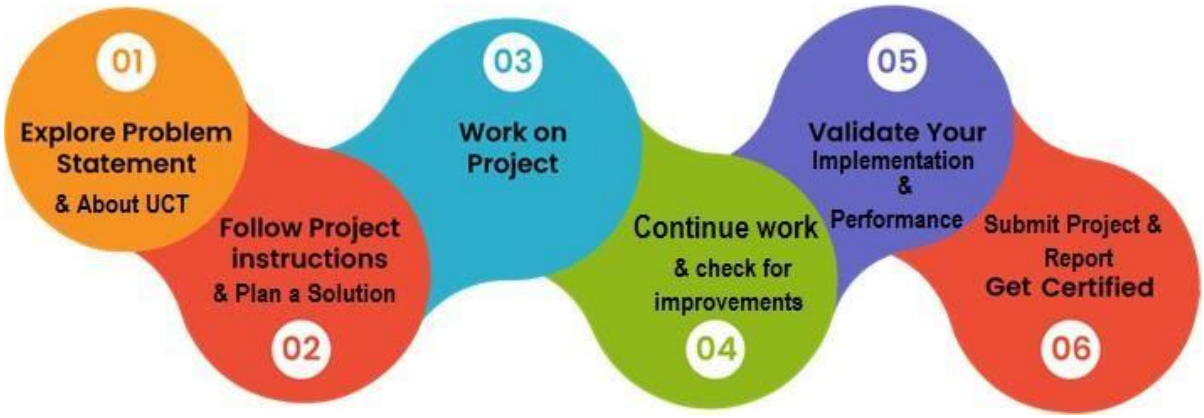
Summary of the whole 6 weeks' work.

About need of relevant Internship in career development.

Brief about Your project/problem statement.

Opportunity given by USC/UCT.

How Program was planned



Your Learnings and overall experience.

Thank to all (with names), who have helped you directly or indirectly.

Your message to your juniors and peers.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

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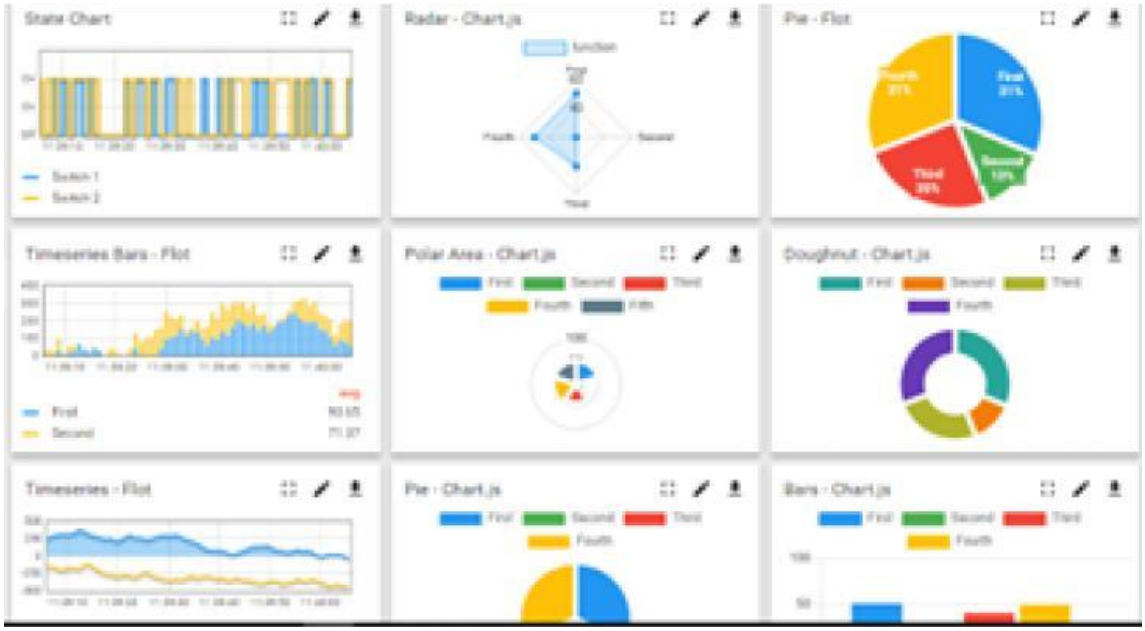
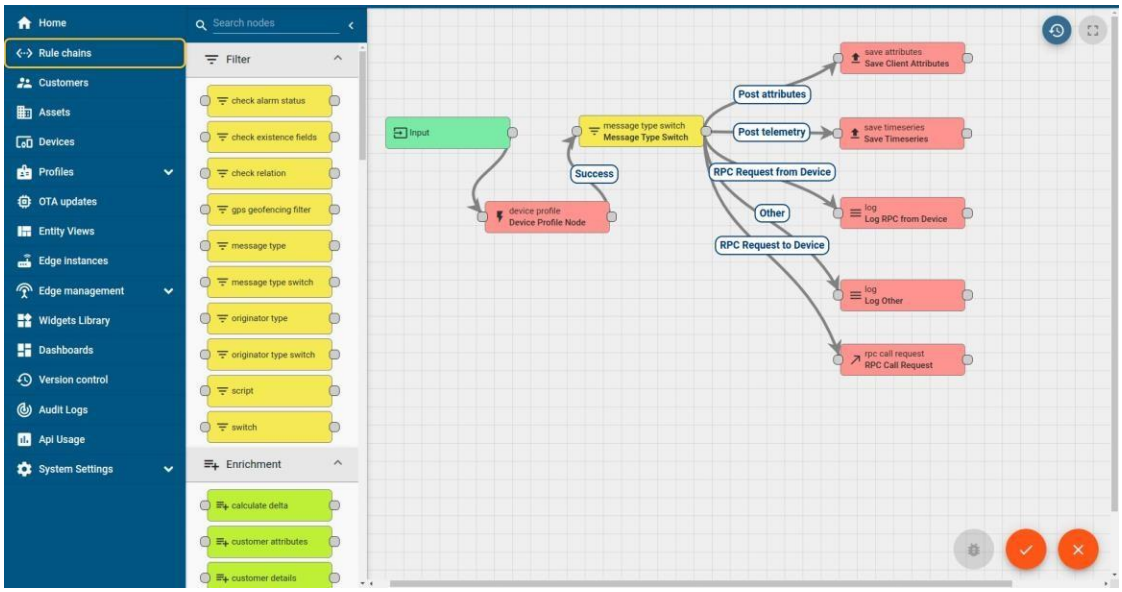
- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine


State Chart
Radar - Chart.js
Pie - Flot
Timeseries Bars - Flot
Polar Area - Chart.js
Doughnut - Chart.js
Timeseries - Flot
Pie - Chart.js
Bars - Chart.js


Home
Rule chains
Customers
Assets
Devices
Profiles
OTA updates
Entity Views
Edge Instances
Edge management
Widgets Library
Dashboards
Version control
Audit Logs
Api Usage
System Settings
Search nodes
Filter
check alarm status
check existence fields
check relation
gps geofencing filter
message type
message type switch
originator type
originator type switch
script
switch
Enrichment
calculate delta
customer attributes
customer details
Input
message type switch
Message Type Switch
Success
device profile
Device Profile Node
Post attributes
Post telemetry
RPC Request from Device
Other
RPC Request to Device
save attributes
Save Client Attributes
save timeseries
Save Timeseries
log
Log RPC from Device
log
Log Other
rpc call request
RPC Call Request

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ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

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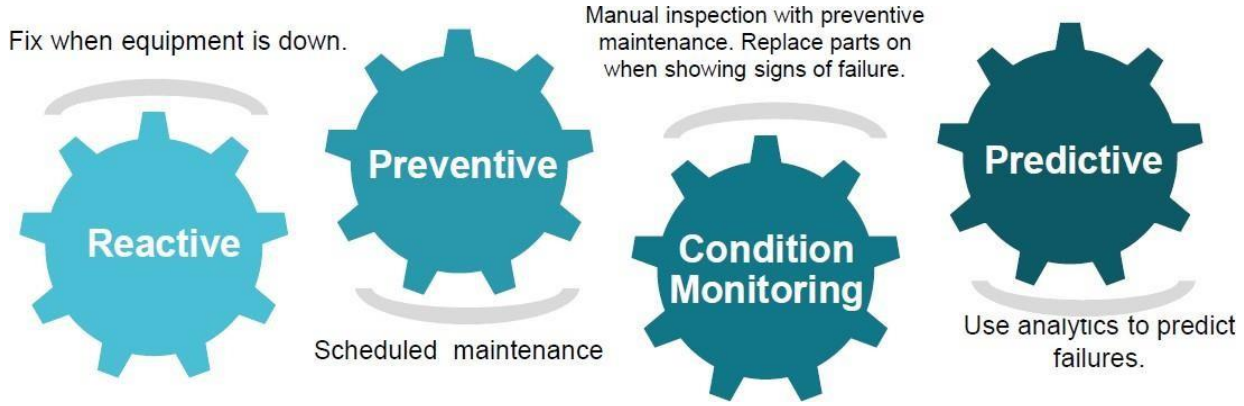
iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc. iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

upSkill CAMPUS

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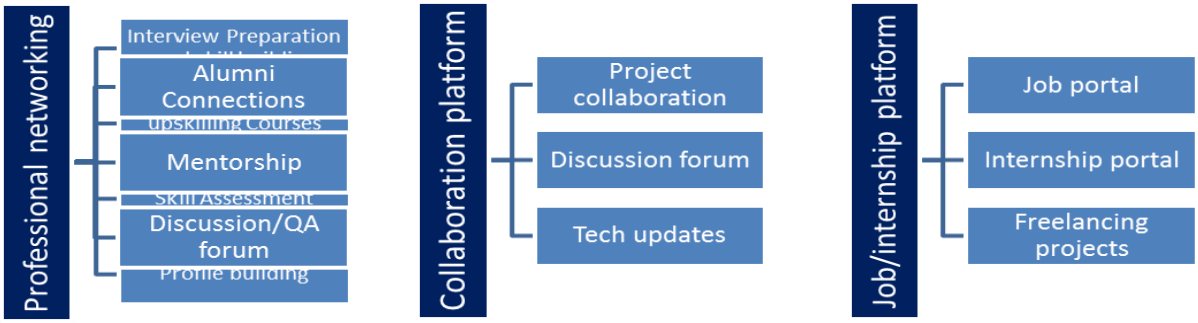
upSkill CAMPUS

Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to ☛ get

practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] upSkill Campus Internship Program Guidelines

[2] UCT Official Website and Project Problem Statement

[3] Python Documentation

[4] Scikit-learn Documentation

2.6 Glossary

Terms	Acronym
ML	Machine Learning
EDA	Exploratory Data Analysis
CSV	Comma Separated Values
R2 Score	Coefficient of Determination
MSE	Mean Squared Error
Random Forest	Ensemble Learning Algorithm

3 Problem Statement

Agriculture plays a vital role in the Indian economy and serves as a primary source of livelihood for a large section of the population. Accurate prediction of crop production is essential for farmers, government authorities, and agricultural industries to make informed decisions related to planning, storage, distribution, pricing, and resource management.

Crop production depends on multiple factors such as crop type, state, season, and cost of cultivation. Due to the complexity and variability of these factors, traditional methods of estimating crop production, which are often based on manual calculations or past experience, may not always provide accurate and reliable results.

Therefore, there is a need for an efficient and automated system that can analyze historical agricultural data and generate accurate predictions.

The problem statement of this project is:

"To develop a machine learning-based system that predicts agricultural crop production in India using historical data and input features such as crop type, state, season, and cost of cultivation."

The goal of this project is to build a regression-based model that can learn patterns from past data and provide reliable predictions of crop production, thereby assisting in better agricultural planning and decision-making.

4 Existing and Proposed solution

#Summary of existing solutions provided by others, what are their limitations?

In many rural and semi-urban areas, farmers and agricultural organizations estimate crop production using traditional methods. These methods are mostly dependent on:

- Manual calculations
- Previous year production records
- Local knowledge and farmer experience
- Seasonal assumptions and general estimations

In some cases, basic statistical techniques such as average production calculation are used. However, these approaches have several limitations:

- Predictions are often inaccurate due to lack of proper data analysis
- Manual estimation is time-consuming and requires continuous human effort

- Traditional methods cannot efficiently handle large datasets
- Variations across different states and seasons are difficult to analyze manually
- These methods do not provide real-time or quick decision-making support

Due to these limitations, there is a need for a more efficient and intelligent system that can provide accurate predictions using modern data-driven techniques.

The proposed solution is to develop a machine learning-based system that predicts agricultural crop production in India using historical data.

The system uses a regression-based model to analyze past agricultural data and identify patterns between different input features and crop production. Based on these patterns, the model can predict future crop production values.

In this project, the dataset includes features such as:

- Crop name
- State
- Season
- Area / Quantity
- Cost
- Production (Target Variable)

The system performs data preprocessing, converts categorical data into numerical format, and trains a Linear Regression model to make predictions.

The proposed system provides several advantages:

- Improved prediction accuracy compared to traditional methods
- Ability to handle large datasets efficiently
- Faster and automated prediction process
- Better analysis of state-wise and seasonal variations

The system performs the following tasks:

- Loads the dataset in CSV format
- Performs data cleaning by removing missing values and duplicate records

- Performs Exploratory Data Analysis (EDA) using graphs to understand patterns
- Applies feature engineering techniques such as label encoding for categorical columns
- Splits the dataset into training and testing data
- Trains a regression model (Random Forest Regressor / Linear Regression)
- Evaluates model performance using metrics such as R2 Score and MSE
- Provides a Streamlit-based user interface for prediction

#What value addition are you planning?

The proposed system provides the following advantages:

- Automated crop production prediction
- Faster and more accurate results compared to manual estimation
- Helps farmers and organizations make better decisions
- Can be improved in future by adding weather and soil-related parameters

Thus, the proposed machine learning-based solution provides an efficient and scalable method for crop production prediction.

4.1 Code submission (Github link) : https://github.com/Gaurav-21-06/Crop-Production-prediction

4.2 Report submission (Github link) : https://github.com/Gaurav-21-06/Crop-Production-prediction/blob/main/Project_Report.md

4.3 Comparison of Existing and Proposed Solution

The existing system for predicting crop production mainly relies on traditional methods such as manual calculations, past records, and farmer experience. These methods are simple but often lack accuracy and are time-consuming. They also fail to handle large datasets and multiple influencing factors effectively.

In contrast, the proposed system uses machine learning techniques to analyze historical agricultural data and predict crop production more accurately. It considers multiple factors such as crop type,

state, season, and cost of cultivation. The system is automated, faster, and capable of providing reliable predictions, making it more efficient for real-world applications.

Parameter	Existing System	Proposed system
Method	Manual / Traditional	Machine Learning Model
Accuracy	Low	High
Speed	Slow	Fast
Data Handling	Limited	Handles Large Data
Automation	Available	Fully Automated

5 Proposed Design/ Model

The proposed system is designed to predict agricultural crop production using a machine learning approach. The system follows a structured pipeline that includes data collection, preprocessing, model training, and prediction.

Initially, the dataset containing agricultural information is collected. This dataset includes features such as crop type, state, season, and cost of cultivation. The collected data is then preprocessed by handling missing values and converting categorical data into numerical format using encoding techniques.

After preprocessing, the dataset is divided into training and testing sets. A Linear Regression model is then trained on the training data to learn the relationship between input features and crop production.

Once the model is trained, it is used to predict crop production based on user inputs. The system takes input values, processes them, and generates predicted output. This prediction helps in better decision-making and planning in the agricultural sector.

The entire system is implemented using Python and relevant machine learning libraries, and the final model can be deployed using a web interface.

5.1 High Level Diagram

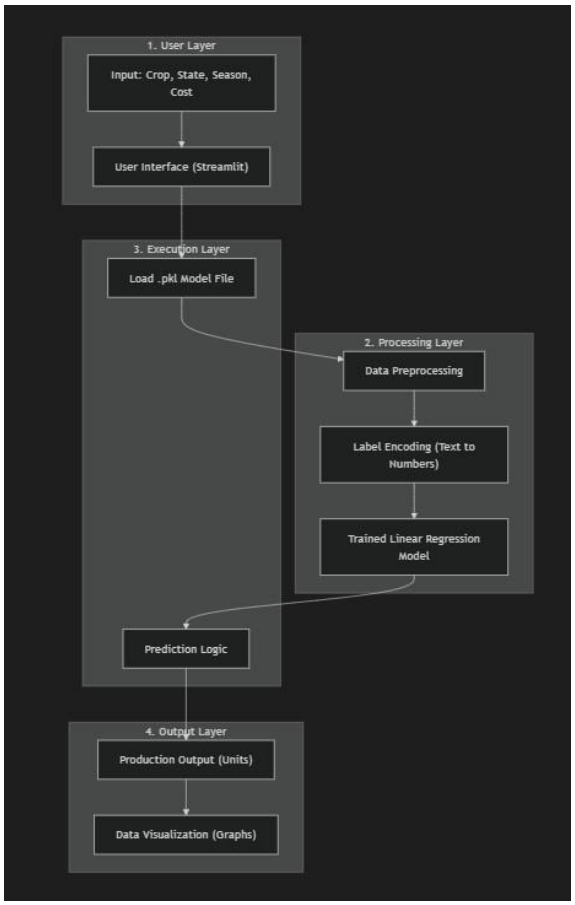


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

High Level Flow:

1. **Upload Dataset**:
   - You load the agriculture_data.csv file using the pandas library (pd.read_csv).
2. **Data Cleaning**:
   - You check for empty cells (isnull()) and ensure the data is "clean" so the AI doesn't get confused by missing information.
3. **EDA (Exploratory Data Analysis) Visualization**:
   - You create graphs (like Scatter Plots) to see how the cost of cultivation relates to the production value.
4. **Feature Encoding**:
   - You use LabelEncoder to turn text labels (like 'Punjab', 'Maize') into numbers (0, 1, 2...) so the mathematical model can understand them.

5. **Train-Test Split**:
    - You divide your data into two parts: Training (80%) to teach the AI, and Testing (20%) to see if it can predict accurately on data it hasn't seen before.
6. **Model Training**:
    - You feed the training data into the Linear Regression model (model.fit). This is where the AI "learns" the patterns.
7. **Model Evaluation**:
    - You compare the Actual values from your test set with the Predicted values from your model to see how well it performed.
8. **Prediction Output**:
    - Finally, you use the trained model to take a New Input (from your website) and output a specific Production Value in Units.

5.2 Low Level Diagram

Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

Low Level Steps:

- Import required libraries (pandas, numpy, scikit-learn)
- Load the agricultural dataset (CSV file)
- Handle missing values and remove duplicate records
- Perform basic data analysis and visualization (if required)

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- Convert categorical features (Crop, State, Season) into numerical format using Label Encoding
- Select input features (Crop, State, Season, Cost) and target variable (Production)
- Split the dataset into training and testing sets
- Train the Linear Regression model on training data
- Evaluate the model using metrics such as R2 Score and Mean Squared Error (MSE)
- Save the trained model as a .pkl file

5.3 Interfaces (if applicable)

The project uses **Streamlit** as the user interface. The Streamlit app includes:

- Dataset Upload option
- Dataset Preview
- Graphs for analysis
- Model training and evaluation output
- Prediction form

6 Performance Test

Constraints Identification (Industrial Perspective)

In real industries, machine learning models must work efficiently with large datasets and should provide accurate predictions within limited time and computing resources. While developing this crop production prediction system, the following major constraints were identified:

1. Dataset Quality Constraint

Agricultural datasets often contain missing values, duplicate records, and inconsistent data entries due to manual data collection and storage processes. These issues can negatively affect the performance of the machine learning model and lead to inaccurate predictions.

How it was handled:

This constraint was addressed by applying data preprocessing techniques such as removing null values, eliminating duplicate records, and ensuring data consistency before training the model.

Test Result:
After data cleaning, the dataset became more structured and reliable. The model was trained successfully without errors, and prediction accuracy improved.

2. Categorical Data Handling Constraint

The dataset includes categorical features such as crop type, state, and season, which cannot be directly used in machine learning models.

How it was handled:

Label Encoding technique was applied to convert categorical data into numerical format so that it can be used for model training.

Test Result:
After encoding, the data was successfully converted into a suitable format, and the model was able to process it effectively.

3. Model Accuracy Constraint

Achieving good prediction accuracy is important, as inaccurate predictions can affect decision-making in agriculture.

How it was handled:

A Linear Regression model was used and evaluated using performance metrics such as R2 Score and Mean Squared Error (MSE).

Test Result:
The model achieved satisfactory accuracy, and the predictions were reasonably close to actual values, making it suitable for basic agricultural prediction tasks.

4. Computational Efficiency Constraint

In real-world applications, crop production datasets can increase in size over time and may include additional factors like rainfall, soil type, temperature, and fertilizer usage.

How it was handled:
The project design is modular (separate model.py and app.py), making it scalable and easy to update with new features and models.

Recommendation:
For large industrial datasets, cloud platforms like AWS/Azure can be used to improve computation speed and scalability.

The identified constraints were addressed through data preprocessing, selection of suitable machine learning models, evaluation metrics, and modular project architecture. These steps ensure the project is suitable for real-world industrial applications and not limited to academic implementation.

6.1 Test Plan/ Test Cases

Test Case	Input	Expected Output
Load Dataset	Agriculture CSV dataset	Datasets loads successfully
Data Cleaning	Dataset with null values	Null values removed
Data Encoding	Crop column	Converted into numeric format
Data Splitting	Preprocessed dataset	Data split into training and testing sets
Model Training	Training dataset	Model trains successfully without error
Model Evaluation	Test dataset	R2 score and MSE generated
Prediction	User inputs crop/state/season	Predicted crop production output

6.2 Test Procedure

- Ran the application using Flask (app.py).
- Loaded the agricultural dataset (CSV file) into the system.
- Verified that the dataset was loaded correctly by displaying initial records.
- Performed data cleaning by removing null values and duplicate entries.
- Conducted basic data analysis and visualization (if applicable).
- Applied Label Encoding to convert categorical features (Crop, State, Season) into numerical format.
- Selected input features (Crop, State, Season, Cost) and target variable (Production).
- Split the dataset into training and testing sets (80/20 ratio).
- Trained the model using Linear Regression algorithm.
- Evaluated the model using performance metrics such as R2 Score and Mean Squared Error (MSE).
- Tested the prediction system by entering sample user inputs and verified the predicted crop production output

6.3 Performance Outcome

The machine learning model was evaluated using regression evaluation metrics to measure its performance and prediction accuracy.

- R2 Score: It indicates how well the model explains the variance in crop production based on the input features such as crop type, state, season, and cost.

- Mean Squared Error (MSE): It measures the average squared difference between the actual and predicted crop production values. A lower MSE indicates better model performance.

The results were analyzed and displayed using:

- R2 Score value
- MSE value
- Comparison between actual and predicted values (graphical representation)

The Linear Regression model successfully predicted crop production values based on the given input parameters. The predicted results were reasonably close to the actual values, indicating that the model performs effectively for basic tasks. .

7 My learnings

During this internship, I gained practical knowledge of developing a complete machine learning project from scratch. The experience helped me understand both theoretical concepts and their real-world implementation. The key learnings from this project are as follows:

- Learned how to work with real-world agricultural datasets and understand their structure.

- Improved skills in data preprocessing, including handling missing values and removing duplicate records.

- Gained experience in performing basic data analysis and understanding patterns in the dataset.

- Understood the importance of encoding categorical variables such as crop, state, and season for machine learning models.

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- Learned how to split the dataset into training and testing sets for proper model evaluation.
- Implemented a Linear Regression model for predicting crop production.
- Learned evaluation techniques using R2 Score and Mean Squared Error (MSE).
- Gained hands-on experience in building a web-based application using Flask for user interaction.
- Understood the process of saving and loading trained models using .pkl files.
- Improved technical skills in Python libraries such as Pandas, NumPy, and Scikit-learn.
- Learned the importance of using GitHub for version control and project management.

Overall, this internship enhanced my confidence in machine learning and helped me understand how realworld industrial projects are designed, developed, and deployed.

8 Future work scope

The current system provides a basic machine learning solution for predicting agricultural crop production. However, there are several areas where the project can be further improved and expanded in the future.

- Advanced machine learning algorithms such as Random Forest, Decision Trees, or Neural Networks can be implemented to improve prediction accuracy.
- Additional input features such as rainfall, temperature, soil type, and humidity can be included to make the model more accurate and realistic.
- Real-time data integration can be added using APIs to provide up-to-date predictions.
- The user interface can be enhanced to make the system more interactive and user-friendly.
- The system can be deployed on cloud platforms for better accessibility and scalability.
- Visualization dashboards can be added to display trends and insights in graphical form.

- The model can be optimized for better performance and faster predictions.

- A mobile application version of the system can be developed for easier access by farmers.

Overall, the project has strong potential for further development and can be enhanced into a more advanced and practical agricultural decision-support system. These improvements will make the project more scalable and suitable for real-world agriculture industries.